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Free LLMs for genomic analysis orchestration in a typical lab

Introduction

Frontier models from Anthropic, OpenAI, and Google write quality working code for data analysis tasks, but they cost cents to dollars per call, resulting in rapidly ballooning bills. It really does not make sense asking Opus to write a hundred-line bash script every time a few FASTQ files need to be aligned to reference wasting a budget that should go to harder problems. Can we instead ask an expensive frontier model to write a recipe once, and then have a free, small open-weight model running on the lab's own hardware turn that recipe into a working script every time after?

Prior evaluations of large language models (LLMs) in bioinformatics fall into three categories that differ in task granularity, model coverage, and compute environment. The earliest studies measured factual recall on genomics question-and-answer benchmarks: GeneGPT [22] modified a closed-source model with live access to the NCBI web APIs and reported strong scores on the GeneTuring benchmark, and the 2025 GeneTuring re-evaluation [23] extended this design to 1,600 questions across sixteen model configurations spanning closed frontier systems (GPT-4o, Claude 3.5, Gemini) and small biomedical models (BioGPT, BioMedLM). A second category of studies shifted from factual recall to code generation on isolated tasks. BioCoder [24] graded 2,269 short bioinformatics functions against unit tests and compared open- and closed-source models, but its open-source arm was restricted to code-completion models that pre-date the current generation of open-weight chat models — publicly released model weights that anyone can download and run locally. BioLLMBench [25] scored GPT-4, Gemini, and LLaMA on 24 tasks and found LLaMA frequently failed to emit runnable code, while an evaluation of 104 Rosalind problems [26] placed GPT-3.5 ahead of GPT-4o and Llama-3-70B in pass rate and compared all three to human solvers. The third category of studies targets end-to-end pipelines and agentic systems — LLMs that plan, call tools, and revise their own output in a loop. BixBench [27] grades closed-source models on more than 50 Dockerised computational-biology scenarios and reports approximately 17 % open-answer accuracy; BioMaster [28] wraps a Plan/Task/Debug/Check loop around retrieval-augmented generation (RAG, the practice of supplying relevant documentation to the model at query time) for RNA-seq, ChIP-seq, scRNA-seq, and Hi-C analyses; BioAgents [29] fine-tunes small open-weight models for local execution and reports expert-level performance on conceptual tasks; two workflow-language studies generate Galaxy and Nextflow pipelines using closed-source models combined with DeepSeek-V3 [30] or rerank Galaxy workflows with Mistral-7B against GPT-4o [31].

Our study is different. We test whether a recipe authored once by a frontier model can be executed end-to-end by a free, locally-runnable open-weight implementer at frontier accuracy. Our goal is to characterize the recipe-implementer split on a real bioinformatics task — per-sample mtDNA variant calling on four paired-end Illumina samples — across the lab-scale hardware tier (NVIDIA Jetson AGX Orin, RTX 5080 desktop, MacBook Pro M4 Pro, and 2× RTX A5000 workstation). Briefly, claude-opus-4-7 authors a series of plans of increasing detail; six 2026-release open-weight implementers run those plans on a desktop GPU; the strongest implementer is then cross-platform validated and stress-tested with a programmatic error-injection.

To decide which open models to use we first need to survey the landscape of available models. For 2026 this landscape is summarized in Table 1 below.

Table 1. Major open-weight model families as of May 2026.

Lab	Family / latest	Sizes (total; active for MoE)	Architecture	Coder variant	Reasoning variant	License
Meta	Llama 4 [1]	Scout 109 B (17 B-A); Maverick 400 B (17 B-A); Behemoth ~2 T (unreleased)	Sparse MoE; multimodal	—	—	Llama 4 Community License (700 M-MAU cap; excludes EU users)
Alibaba	Qwen3.6 [2]	0.6 B → 397 B (17 B-A); 27 B dense	Dense + MoE; thinking toggle	Qwen3-Coder-Next	Qwen3 thinking mode	Apache 2.0
DeepSeek	DeepSeek-V4 [3]	V4-Flash 284 B (13 B-A); V4-Pro 1.6 T (49 B-A)	Sparse MoE	DeepSeek-Coder	V4-Pro Max mode	MIT
Mistral	Mistral Large 3, Mixtral, Codestral, Ministral [4]	3 B → 675 B (41 B-A)	Granular MoE flagship; dense small	Codestral 25.08	—	Apache 2.0 (Codestral non-production)
Google	Gemma 4 [5]	E2B, E4B, 26 B-MoE, 31 B-dense	Dense + MoE; multimodal	—	—	Apache 2.0
IBM	Granite 4.1 / 4.0 [6]	350 M, 1 B, 3 B, 8 B, 30 B; 4.0-H-Small 32 B (9 B-A)	Dense (4.1); hybrid Mamba-2 / MoE (4.0)	Granite-Code	—	Apache 2.0
Allen AI	OLMo 3 [7]	7 B, 32 B	Dense	—	OLMo 3-Think	Apache 2.0; **fully open** (weights + data + recipes)
Microsoft	Phi-4 [8]	mini 3.8 B, multimodal 5.6 B, 14 B	Dense; multimodal	—	Phi-4-reasoning 14 B	MIT

Lab	Family / latest	Sizes (total; active for MoE)	Architecture	Coder variant	Reasoning variant	License
NVIDIA	Nemotron-3 *(new in late 2025)* [9]	Nano-Omni 30 B (3 B-A); Super 120 B (12 B-A)	Hybrid Mamba-Transformer MoE	—	built-in agentic stack	NVIDIA Open Model License; data + recipes
Cohere	Command A, Aya [10]	Aya 3.35 B → Command A 111 B	Dense	—	—	CC-BY-NC 4.0 — **research only**

Legend.

- **Architecture.** Modern language models read input one piece at a time (each piece, called a *token*, is a short word or word-fragment) and predict the next piece, by passing each token through a set of trainable mathematical operations. Three architectural styles appear in the table:
  - *Dense.* Every internal weight is used on every token — the original transformer design and what most pre-2024 LLMs were. Simple to deploy, but compute per token scales with the model's full size.
  - *Sparse Mixture-of-Experts (MoE).* The model is split into many parallel "expert" sub-networks plus a small "router" that, for each token, picks one or two experts to handle it; the other experts stay idle. This lets the model be very large in *total* parameters (high stored capacity) while spending the per-token compute of a much smaller model. The cost is memory: every expert still has to be loaded, even the ones that won't fire on a given input. Every flagship released since late 2025 uses this design.
  - *Hybrid Mamba-Transformer.* A newer variant that replaces some standard layers with *state-space* (Mamba) layers. The practical consequence is that compute on a long input grows linearly with input length rather than quadratically — relevant now that flagships accept inputs of 500,000 to 1,000,000 tokens (a small book).
  - *Multimodal.* Accepts images, and sometimes audio, alongside text.
- **Parameters (B = billion, T = trillion).** Parameters are the trained numerical weights inside a model — loosely analogous to synapse strengths. More parameters mean more stored capacity but also more memory and more arithmetic per token. Two numbers matter:
  - *Total* parameters set disk and GPU-memory cost. A 70 B-parameter model occupies roughly 35–40 GB of memory at the standard 4-bit compression used for inference (a 16-bit "full precision" version would need ~140 GB).
  - *Active* parameters (notation `17 B-A` = 17 billion active per token) set the per-token compute cost. For dense models the two numbers are equal. For MoE models the active count is much smaller than the total because most experts don't fire on a given token. A 109 B (17 B-A) MoE runs about as fast per token as a 17 B dense model, but the machine still has to keep all 109 B parameters loaded.
- **Coder variant.** A general model further trained on a large corpus of source code (and tested on standard code-task benchmarks). Coder variants beat their general siblings on bash, Python, and refactor tasks but lose on chat, math, and translation.
- **Reasoning variant.** A model that, before answering, writes out a step-by-step "chain of thought" that the user typically does not see; training rewards the model when its final answer is verifiably correct. Pioneered by DeepSeek-R1 (January 2025); every major lab has since copied the recipe. Reasoning variants score higher on math, code, and multi-step problems but take roughly 5×–50× longer per call (and cost 5×–50× more).

Several models are omitted from Table 1: Apple's 3-billion-parameter on-device model ships only inside iOS and macOS 26, and xAI has openly released only Grok-1 (March 2024) and Grok-2.5 (August 2025) — everything newer is closed. Every flagship since late 2025 is a sparse MoE, but dense persists below ~40 B because it is simpler to deploy.

One practical reason narrow the choice for most biomedical labs – hardware required: the four size tiers in Table 1 map to four very different machines, from a \$400–\$600 consumer card for the smallest models to a multi-GPU server costing several hundred thousand dollars for the trillion-parameter tier (Table 2).

Table 2. GPU options and ballpark May 2026 US street prices for running each Table 1 model class locally at 4-bit compression. Where multiple cards are needed, the listed price is the per-card cost.

Model class (Table 1 example)	GPU memory needed	GPU options (May 2026 USD)	Refs
7–8 B dense (Phi-4, Granite 8B, OLMo 3 7B)	~5–6 GB	NVIDIA RTX 5060 Ti 16 GB (~\$560); RTX 4060 Ti 16 GB (~\$430); RTX 5070 12 GB (~\$635); Apple M4 Mac mini 16 GB unified (~\$600)	[11, 16]
27–32 B dense (Qwen3.6-27B, Gemma 4 31B, OLMo 3 32B)	~17–22 GB	NVIDIA RTX 5090 32 GB (~\$2,900–\$3,500, street price 50–75 % over \$1,999 MSRP); RTX 4090 24 GB (~\$1,500–\$2,200, EOL Oct 2024); RTX A5000 24 GB (~\$700–\$1,400 used); Apple Mac Studio M3 Ultra 96 GB unified (~\$4,000)	[11, 12, 16]
100–400 B MoE (Llama 4 Scout 109B, Maverick 400B)	~55–250 GB	NVIDIA RTX Pro 6000 Blackwell 96 GB (~\$8,500); 2× RTX 5090 (~\$6,000–\$7,000 total); RTX 6000 Ada 48 GB (~\$6,800); H100 80 GB (~\$25K–\$33K); AMD Instinct MI300X 192 GB (~\$15K–\$20K, OEM-only); Apple Mac Studio M3 Ultra 256 GB unified (~\$9,500)	[11, 12, 13, 14, 16]
Trillion-parameter MoE (DeepSeek-V4-Pro 1.6T)	~700+ GB	NVIDIA H200 141 GB (~\$31K–\$40K each); B200 192 GB (~\$35K–\$55K each); 8-GPU DGX H200 server (~\$350K–\$500K); GB200 NVL72 rack (~\$3M+); cloud rental on B200 ~\$2.25–\$16/GPU-hour	[13, 15]

Note: May 2026 prices are dominated by an ongoing HBM/GDDR7 shortage; consumer NVIDIA 50-series cards and high-RAM Apple Mac Studio configurations currently sell 30–75 % above launch MSRP.

Given this (rapidly evolving) landscape we decided to do the following experiment: take a common sequencing data processing workflow and ask open models running on hardware accessible to an average research lab to design and execute the analysis. In doing so we experimented with a range of possibilities ranging from allowing open models to figure out everything by themselves to guiding them using a very detailed plan produced by commercial frontier models. We further complicated these tasks by simulating a variety of errors that may occur during workflow execution.

Materials and Methods

Hardware

We assembled five computers spanning the lab-scale hardware tier (Table 3) — a salvaged workstation, a gaming-GPU desktop, two recent Apple-silicon MacBooks, and the NVIDIA Jetson AGX Orin developer kit. The Jetson is a "RaspberryPi"-like NVIDIA offering that costs under \$2,000 and has a small footprint, making it a lab-ready tiny-but-powerful workstation.

Table 3. Test machines used in this study.

Computer	Manufacturer	Year released	RAM	OS	GPU
NVIDIA Jetson AGX Orin Developer Kit	NVIDIA	2022	64 GB LPDDR5 (unified with GPU)	Ubuntu 22.04 LTS (NVIDIA JetPack 6)	Integrated Ampere, 2,048 CUDA cores + 64 Tensor cores
RTX 5080 desktop	Dell	2025 (GPU)	128 GB DDR5 (system)	Linux (Ubuntu)	NVIDIA RTX 5080, 10,752 CUDA cores, 16 GB GDDR7
MacBook Pro M4 Pro (48 GB)	Apple	2024	48 GB LPDDR5X (unified with GPU)	macOS Sequoia 15.6	Apple M4 Pro integrated GPU, 16 to 20 cores
2× NVIDIA RTX A5000 desktop	Dell	2021 (GPUs)	256 GB DDR4 (system)	Linux (Ubuntu 25.10)	2× NVIDIA RTX A5000, 8,192 CUDA cores each, 24 GB GDDR6 each (48 GB total)

System RAM listed for the two Linux desktops reflects the build configuration; for inference workloads the relevant memory is the GPU VRAM (last column). For the Jetson and the MacBook, RAM is unified between CPU and GPU and the model can use up to roughly the listed RAM minus the operating-system reservation.

A model's GPU-memory footprint scales with its parameter count: at the 4-bit quantization used for all local runs here, roughly 0.5 GB per billion parameters. The plan-granularity-gradient sweep was executed on the **RTX 5080 desktop** with six 2026-release open-weight implementers — `gemma4:26b`, `gemma4:e4b`, `glm-4.7-flash`, `gpt-oss:20b`, `qwen3.6:27b`, and `qwen3.6:35b-a3b` — chosen so the family covers small dense (4 B effective), mid-size dense (26–27 B), and small-active sparse-MoE architectures within or just above the 5080's 16 GB VRAM ceiling at 4-bit. The strongest implementer (`qwen3.6:27b`, 17 GB at 4-bit) is then cross-platform validated on the Jetson AGX Orin (64 GB unified memory), MacBook Pro M4 Pro (48 GB unified memory), and 2× RTX A5000 (48 GB total VRAM) — all three host the model comfortably. The Anthropic API models (`claude-opus-4-7` as plan author throughout, plus `claude-sonnet-4-6` and `claude-haiku-4-5` as confirmation implementers) run remotely and are independent of host GPU.

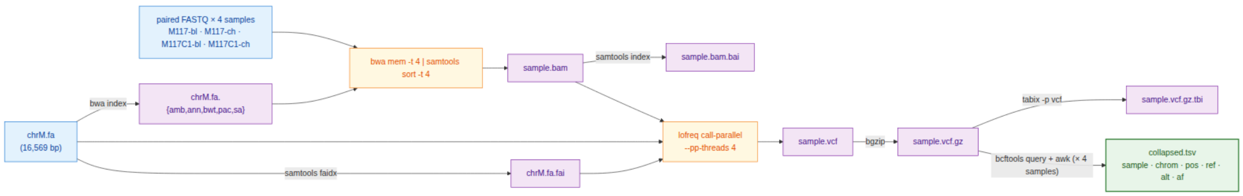
Data

We selected a small dataset [17] derived from our previous work on the analysis of mutational patterns in human mitochondria [18]. It contains four paired-end Illumina samples derived from blood and cheek tissues of a mother–child pair. It contains two fixed changes and a low-frequency variant in the child's cheek sample — a heteroplasmy example.

Workflow

We developed a simplified version of a haploid variant-calling workflow (original: [19]) that omits pre-processing steps before variant calling and the variant-annotation phase [20, 21]. The structure of the workflow is shown in Fig. 1 below.

Figure 1. Data flow of the simplified mtDNA variant-calling workflow used in this study. Per-sample steps (alignment through tabix) run independently for each of the four samples; the only inter-sample dependency is the final `bcftools query + awk` fan-in that builds `collapsed.tsv`.



Plans

We asked the free implementers to perform variant calling under conditions of varying recipe detail — from no plan at all to a very-detailed step-by-step specification (Table 4; full text in the Supplement). Every plan was authored by claude-opus-4-7 in the planner role of the recipe-implementer split. Briefly, Opus wrote v1 (the lean reference recipe); v2 (a hyper-detailed recipe authored in response to v1's poor implementer performance — see Results); v1.25 and v1.5 (controls that localize which part of v2 carries the score improvement); v1g (a robustness check authored mechanically from the Galaxy IUC tool registry rather than by Opus); v0.5 (a no-recipe condition with only the tool order); and v2_defensive (v2 plus an error-handling skeleton).

Each model is run under one of two conditions, which we call tracks. In Track A, the model receives a written plan (one of the rows of Table 4 with a non-empty *File* column) together with the tool inventory and writes `run.sh` from the plan. In Track B, the model receives the problem statement and the tool inventory only — no plan — and writes `run.sh` from scratch. Track B exists to bound how much of the implementer's behaviour comes from the model's prior training versus from the supplied recipe; every other plan in Table 4 is a Track A condition.

Table 4. Plan variants used in this study. Plans of increasing granularity, plus two "no-plan" controls. File sizes are bytes; full text for each plan file is in the Supplement.

Plan	File	Size	Short summary	Hypothesis tested
Track B	*(no plan)*	0	Problem statement + tool inventory only; no implementation hints	How much can a model do from scratch?
v0.5	`prompts/track_b_with_order.tmpl`	1.4 KB	Track B + one line giving the tool order: `bwa` → `samtools` → `lofreq` → `bcftools` → `awk`	Does sequencing alone help local models?
v1 (lean)	`plan/PLAN_v1.md`	3.1 KB	Numbered bullets naming tools and key flags; no exact command lines	Reference lean plan
v1.25	`plan/PLAN_v1p25.md`	3.1 KB	v1 + the exact `lofreq call-parallel` command line	Does one full command bridge the v1 → v2 gap?
v1.5	`plan/PLAN_v1p5.md`	1.3 KB	v2 with every prose paragraph and Gotchas block stripped — code fences only	Are the prose explanations load-bearing or decorative?
v1g	`plan/PLAN_v1g.md`	4.2 KB	v1 + a Galaxy-IUC-mechanical `lofreq` snippet (extracted from `tools-iuc@39e7456`)	Can a tool registry replace a human plan author?
v2 (detailed)	`plan/PLAN.md`	4.6 KB	Every step gives the exact command line plus Gotchas block	Reference detailed plan

Plan	File	Size	Short summary	Hypothesis tested
v2_defensive	plan/PLAN_v2_defensive.md	6.5 KB	v2 + `try()` helper, output validation after every step, retry-once, per-sample isolation, structured failure log	Does explicit error-handling prose make implementer scripts defensive against runtime tool failures?

Error simulation

To test how each model handles tool failures, we wrapped `bwa` and `lofreq` in short shell scripts ("shims") that imitate the real binaries on disk. Before each run, the harness writes the two shims into a per-run directory and adds that directory to the front of `PATH` — the colon-separated list of directories the shell consults when it needs to find an executable. When the script types `bwa mem`, the shell finds the shim first and runs it instead of the real `bwa`. Each shim reads two environment variables: `EVAL_INJECT_PATTERN` names the failure type and `EVAL_INJECT_TARGET` names which of `bwa` or `lofreq` the failure applies to. For every untargeted invocation the shim passes the call straight through to the real binary; for the targeted invocation it injects the failure and then either passes through or stops. The model's `run.sh` does not know it is being intercepted.

Seven failure patterns cover the kinds of breakage a real lab pipeline encounters (Table 5). Each pattern is paired with one tool (or both), so a full error-matrix cell is one (model × plan × pattern × target tool × seed) combination. We use seeds 42, 43 and 44, and run every cell on both the `v2` and the `v2_defensive` plan to measure how the defensive prose changes outcomes.

Table 5. Failure patterns injected by the PATH shims into `bwa` and / or `lofreq`.

Pattern	Affected tool	What the shim does
`flake_first_call`	both	First call exits 1; later calls pass through.
`one_sample_fails`	both	Shim exits 1 only when `M117C1-ch` appears in the command line; the other three samples run clean.
`slow_tool`	both	Sleep 30 s, then run the real tool — output identical, just delayed.
`stderr_warning_storm`	both	Dump 200 lines of `WARNING:` text to stderr, then run the real tool — output is correct but the stderr looks alarming.
`missing_lib_error`	both	Exit 127 without invoking the real tool, mimicking a missing shared library (`error while loading shared libraries: libhts.so.3: cannot open shared object file`).
`silent_truncation`	lofreq only	Run the real `lofreq`, then truncate the `-o` output file to zero bytes — `lofreq` returns 0 but the VCF is empty.
`wrong_format_output`	lofreq only	Run the real `lofreq`, then strip every variant line from the VCF; only the header survives — the file still parses, but contains no calls.

Beyond the variant-overlap score (defined in *Scoring* below), we score each error-matrix cell on three error-handling metrics — referred to here as the **handle category**, **recover**, and **diagnose** (fields `m_handle`, `m_recover`, `m_diagnose` in the per-cell score JSON). The handle category classifies the script's overall response as `crash` (exited non-zero with no failure log), `propagate` (the failure passed through to a downstream step), `partial` (the script aborted but recorded the failure structurally), or `recover` (the script completed successfully despite the injection). The recover metric is binary: did the script produce the count of valid VCFs that the failure pattern allows? For `one_sample_fails` the best achievable is 3 (the bad sample's VCF is correctly absent); for `silent_truncation`, `wrong_format_output` and `missing_lib_error` the best is 0 (the script should detect the broken or missing output and skip it); for the remaining three patterns the best is 4. The diagnose metric is binary: did the script announce the failure — through a populated `failures.log`, a summary line on stderr, or a sample-name-and-failure-word pair in its output — or did it stay silent? All three metrics are defined in `score/score_run.py:error_handling`.

Scoring

We score every run on a single primary metric — the **variant-overlap (Jaccard) score** — computed automatically by `score/score_run.py` from the script's outputs and the canonical answer key. The variant-overlap score is the per-sample Jaccard overlap of `(chrom, pos, ref, alt)` between the script's PASS-or-unfiltered records and the answer key, requiring matched allele frequencies within ± 0.02 and averaged over the four samples; it ranges from 0 (no overlap) to 1 (every call correct). We gate the score on script execution: a run whose `bash run.sh` invocation does not return exit 0 within the 600-second wall-clock budget contributes its produced VCFs as-is to the average, and any missing VCF counts as zero overlap on its sample. A model that uses a different but valid tool from the one named in the plan — `bcftools mpileup` in place of `lofreq`, for example — gets credit for the variants it called, not for the tool it chose.

For the error-injection cells (described under *Error simulation*) we add the three error-handling metrics defined above (handle category, recover, diagnose). The per-run `score.json` also records additional bookkeeping fields — token counts, USD cost (Anthropic only — Ollama is free), generation and execution times, file-schema completeness, and shellcheck/idempotency status — that are written for reproducibility but are not analyzed in the Results below.

Results

Six open-weight implementers reach frontier accuracy on the most detailed Opus-authored plan

Every (model × plan) condition reported below was run three times with different random seeds (42, 43, 44). A seed is the integer that controls the model's token-sampling stochasticity — the same prompt-and-seed pair produces reproducible output, while different seeds produce independent samples from the same underlying probability distribution, letting us distinguish robust success (all three seeds pass) from intermittent success (only one or two seeds pass).

Five of six 2026-release open-weight implementers reach mean score 1.000 on the v2 plan, the most detailed Opus-authored recipe. We ran the six implementers on the RTX 5080 against every plan in Table 4 — Track A (with plan) and Track B (no plan) — with three seeds per (model × plan) condition, scoring the variant-overlap as the share of called variants matching the truth set within a 0.02 allele-frequency window (Methods, *Scoring*). The five passing implementers are qwen3.6:27b, qwen3.6:35b-a3b, gemma4:26b, gemma4:e4b, and glm-4.7-flash; the sixth, gpt-oss:20b, reaches mean score 0.67 — a budget effect, since its built-in reasoning mode consumes most of the output-token allowance before the script is finished. The recipe-implementer split is operational on commodity hardware: a frontier-tier author writes the recipe once and a 17 GB open-weight implementer reproduces frontier accuracy at zero per-call cost on a desktop GPU.

Plan detail is the binding constraint — a threshold response repaired by a single line

A single command line carries most of the v1 → v2 score lift. To localize what an implementer needs from the recipe, we constructed plans of increasing detail (Fig. 2). At the lean end, claude-opus-4-7 first wrote v1: a numbered bullet list naming each tool and its key flags but no exact command lines. v1 produces an all-or-none response — qwen3.6:27b and glm-4.7-flash reach score 1.000, but the four other implementers fail at score ≈ 0. The detailed v2 recipe, written in response, recovers all six implementers (subject to the gpt-oss:20b budget effect noted above). To isolate which part of v2 is doing the work, we authored two intermediate plans: **v1.25** = v1 plus the single literal `lofreq call-parallel --pp-threads 4 -f data/ref/chrM.fa -o results/{s}.vcf results/{s}.bam` invocation, and **v1.5** = v2 with every prose paragraph and *Gotchas* block stripped, code fences only. v1.25 reproduces the v2 jump for every dense ≥ 26 B implementer (gemma4:26b, qwen3.6:27b, qwen3.6:35b-a3b); v1.5 does not. The active ingredient is the literal command syntax, not the surrounding explanation. Track B (no plan) and v0.5 (no plan plus the tool order bwa → samtools → lofreq → bcftools → awk) confirm the floor: every open-weight implementer's score lands at ≈ 0 in both no-recipe conditions; supplying tool order without flags or arguments does not lower the threshold.

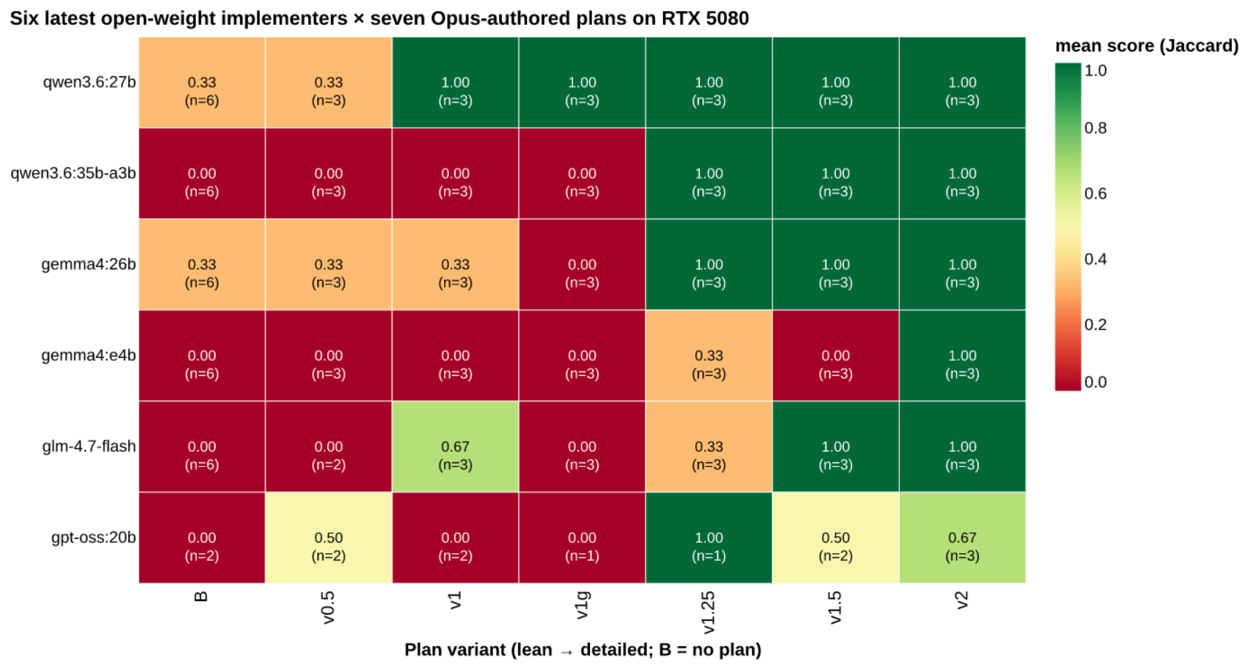


Figure 1: Figure 2. Mean score across three seeds for six open-weight implementers × seven Opus-authored plan variants on the RTX 5080. Columns are ordered by plan detail (left = no plan, right = most detailed). Anthropic API plan-author rows are not shown here because Anthropic models execute on Anthropic's own hardware; they appear in the cross-platform error-injection panel below.

A robustness control rules out the alternative explanation that the v2 effect is specific to Opus's writing style. We constructed **v1g** as v1 plus the lofreq call-parallel snippet extracted mechanically from the Galaxy IUC tool registry (tools-iuc@39e7456), a hand-curated public corpus of bioinformatics tool wrappers. v1g produces a similar implementer lift to v1.25 — the load-bearing element is the command syntax itself, regardless of whether a frontier model or a tool registry is its source.

qwen3.6:27b is the only fully-open Apache-2.0 implementer that clears the lean v1 plan above threshold

Within the six-implementer lineup, two implementers pass v1 with score 1.000 — qwen3.6:27b (Apache 2.0, Alibaba; 17 GB at 4-bit) and glm-4.7-flash (Z.ai; comparable size). qwen3.6:27b is the cleaner choice for a working lab: an Apache-2.0 license with no field-of-use restrictions, a single-file `ollama pull qwen3.6:27b` deployment, and a memory footprint that fits the 24 GB VRAM tier on Table 2 with room to spare. We therefore selected qwen3.6:27b as the protagonist for the cross-platform validation that follows.

Cross-platform: same accuracy, different wall time

Hardware does not limit accuracy on the v2 plan — only wall time. We tested qwen3.6:27b on the three remaining platforms — Jetson AGX Orin, MacBook Pro M4 Pro, and 2× RTX A5000 — under the v2 plan; mean score 1.000 on every platform on every seed. However, wall-clock generation time differs sharply (Table 7). The 2× A5000 returns a generation in under half a minute (median 29 s), the M4 Pro and Jetson take 1.5 to 2 minutes (92 s and 105 s), and the RTX 5080 takes about 5 minutes (302 s). The 5080 row reflects the model partially spilling out of its 16 GB VRAM into system RAM — Ollama handles the spill automatically but slowly. None of the wall times is too slow for a working lab; pick the cheapest box that fits the model in dedicated or unified memory and the score is unchanged.

Table 7. qwen3.6:27b on the v2 plan: median wall-clock time per generation (seconds), inter-quartile range in brackets. score 1.000 on every cell across every platform.

Platform	VRAM/UMA	Median wall time (s)	IQR (Q1–Q3)	n
2× RTX A5000	48 GB total VRAM (fits)	29	[24–31]	36
MacBook Pro M4 Pro	48 GB unified (fits)	92	[91–136]	36
NVIDIA Jetson AGX Orin	64 GB unified (fits)	105	[98–107]	36
RTX 5080 desktop	16 GB VRAM (spills to RAM)	302	[288–322]	6

Defensive scripting under error injection — qwen3.6:27b matches Opus on every platform

qwen3.6:27b matches claude-opus-4-7 cell-for-cell on the error-injection matrix. To probe whether qwen3.6:27b handles the *robustness* case as well as the happy path, we ran the 36-cell error matrix (12 PATH-shim cells × 3 seeds; Methods, *Error simulation*) for both models on three platforms — Jetson, M4 Pro, and 2× A5000 — under both the v2 baseline plan and the v2_defensive plan that adds a try() { ... } helper, per-step output validation, and a structured failures.log (Methods, *Plans*). Both models behave identically at the cell level on every platform tested (Fig. 3, Table 8). On the v2 baseline, both produce 15 recover / 0 partial / 6 propagate / 15 crash; on v2_defensive, both produce 21 recover / 15 partial / 0 propagate / 0 crash — the defensive prose drains every crash into a structurally-recovered partial — and both score mean score 0.625 on the matrix. A 17 GB open-weight implementer matches a frontier-API plan author cell-for-cell and score-for-score on the error matrix.

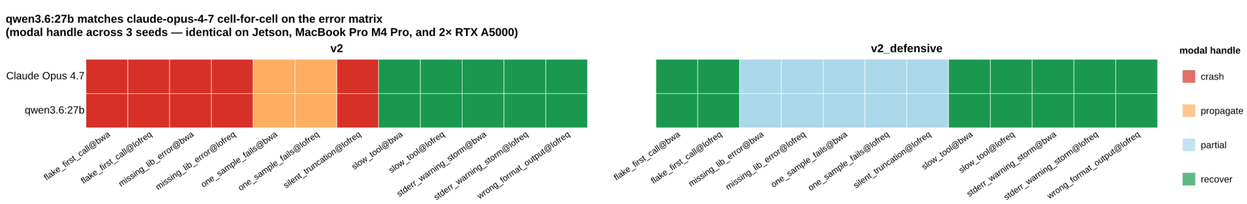


Figure 2: Figure 3. Per-cell modal handle category (recover / partial / propagate / crash) for claude-opus-4-7 and qwen3.6:27b across the 12-cell PATH-shim error matrix. Left panel = v2 baseline plan; right panel = v2_defensive plan. Rows within each panel are the two models. Cells are color-coded by the modal handle category across three seeds; the pattern shown is identical on the Jetson AGX Orin, MacBook Pro M4 Pro, and 2× RTX A5000 platforms (verified for every one of the 48 cells), so a single cross-platform consensus is shown.

Table 8. Per-cell behaviour signature and mean variant-overlap score for qwen3.6:27b and claude-opus-4-7 on the 36-cell error matrix (12 patterns × 3 seeds). Counts and mean score are identical across Jetson, MacBook Pro M4 Pro, and 2× RTX A5000 for every (model, plan) row.

Model	Plan	recover	partial	propagate	crash	mean score
claude-opus-4-7	v2	15	0	6	15	0.333
claude-opus-4-7	v2_defensive	21	15	0	0	0.625
qwen3.6:27b	v2	15	0	6	15	0.333
qwen3.6:27b	v2_defensive	21	15	0	0	0.625

The error-injection result closes the loop on the recipe-implementer architecture: under both the happy-path plan and a defensive plan, a free, locally-runnable Apache-2.0 implementer handles the same workload as the frontier API author at the same accuracy.

Coda — Sonnet and Haiku confirm the implementer role generalizes across the Anthropic family

The implementer role is not specific to Opus within the Anthropic family. We also ran claude-sonnet-4-6 and claude-haiku-4-5 against the same error matrix on the Jetson and 2× A5000 platforms. Both produce the same v2_defensive frontier signature as Opus and qwen3.6:27b — Sonnet at 21 recover / 14 partial / 0 propagate / 1 crash on Jetson and 21/15/0/0 on A5000, mean score 0.583–0.625; Haiku at 21/15/0/0 on both, mean score 0.625. Cross-platform timing for Anthropic models is not informative for hardware-throughput claims — the LLM call is remote — so we report Sonnet and Haiku as confirmation of the implementer role, not as a hardware comparison.

Discussion

The recipe-implementer split works as designed on a real bioinformatics task. A frontier model writes the recipe once, and a free, locally-runnable open-weight model executes it many times at frontier accuracy. The split shifts the cost profile of an LLM-assisted pipeline from cents-to-dollars per call to a one-time recipe-authoring cost plus near-zero per-call inference on the lab's own hardware. For per-sample mtDNA variant calling, qwen3.6:27b — a 17 GB Apache-2.0 model from Alibaba — is the implementer that does the job. It matches Opus cell-for-cell on a 36-cell error-injection matrix on every hardware platform we tested, including the smallest device (NVIDIA Jetson AGX Orin, under $2,000).

A limitation of our study is the single workflow. We measured a four-sample mtDNA variant-calling pipeline on a 16.6 kb reference; the recipe-implementer split may behave differently on larger references, longer pipelines, or analyses that require numerical algorithms beyond shell-stitched binaries (statistics, image processing, or workflows that lack a canonical toolchain). A second limitation is the closed-source plan author. claude-opus-4-7 is an API-only model; reproducibility of the recipes themselves is contingent on Anthropic's continued availability of that model identifier. Recipes can be archived as text — they are short, human-readable, and we include them in the Supplement — but the act of authoring a new recipe for a new task remains tied to the API.

A third limitation concerns the error-injection matrix. The PATH-shim harness probes failure modes that touch tool stdout/stderr, exit codes, and output-file shape; it does not probe failures that depend on tool internals (incorrect alignment scores from a flake in BWA's BWT lookup, for example). Three of seven injection patterns — `missing_lib_error`, `silent_truncation`, and `wrong_format_output` — are unrecoverable in principle on the v2 plan and remain so on v2_defensive; the defensive prose moves these cells from `crash` to `partial` (the script terminates structurally with a populated failures.log) but the actual call cannot proceed. The implementer-quality story is therefore strictly about recoverable failures.

The practical implication for a working lab is direct: a single $400–$600 consumer GPU (RTX 5060 Ti or 5070 with 16 GB VRAM, or a low-cost A4000 with 16 GB VRAM, all listed in Table 2) is enough to run qwen3.6:27b on the lean v1 plan; the same model on a Jetson AGX Orin under $2,000 reproduces frontier accuracy and finishes a per-sample generation in under two minutes. The recipes, scoring scripts, error-injection harness, and aggregate score table are released under MIT on Github (https://github.com/nekrut/LLM-eval-paper) and freely available to anyone with an Internet connection.

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Supplement

Plan files

The full text of each plan file referenced in Table 4. Each plan is reproduced verbatim from `plan/` in the repository and shown inside a fenced Markdown block to preserve its original heading hierarchy and code formatting.

Plan v1 (lean) — `plan/PLAN_v1.md`

```markdown
# Per-sample mtDNA amplicon variant-calling plan

1. **Set globals and prepare results directory**
   - Define `THREADS=4` and the sample list: `M117-bl M117-ch M117C1-bl M117C1-ch`.
   - Create `results/` if missing. Use `set -euo pipefail`.
   - Treat every output step as idempotent: guard each artifact with an existence check (e.g. skip if `

2. **Reference indexing (once, in `data/ref/`)**
   - `samtools faidx data/ref/chrM.fa` → produces `chrM.fa.fai`.
   - `bwa index data/ref/chrM.fa` → produces the `.amb .ann .bwt .pac .sa` set.
   - Skip both if the index files already exist.

3. **Per-sample alignment with `bwa mem`**
   - Use `bwa mem -t 4` with the paired FASTQs `data/raw/{sample}_1.fq.gz` and `data/raw/{sample}_2.fq.
   - Pass the read group via `-R` as a single double-quoted argument containing literal backslash-t bet
     - exact form: `-R "@RG\tID:{sample}\tSM:{sample}\tLB:{sample}\tPL:ILLUMINA"`
     - The `\t` must remain the two characters backslash and `t` - bwa parses them itself. Do NOT use `
     - Separators between key and value are colons `:`, not `=`.

4. **SAM → sorted BAM**
   - Pipe `bwa mem` stdout into `samtools sort -@ 4 -o results/{sample}.bam`.
   - Do NOT run `markdup` or `rmdup`: this is amplicon data where PCR duplicates are expected and biolo
```

5. **BAM indexing**
   - `samtools index -@ 4 results/{sample}.bam` → `results/{sample}.bam.bai`.

6. **Variant calling with `lofreq call-parallel`**
   - Use the `call-parallel` subcommand (not plain `lofreq call`) with `--pp-threads 4`.
   - Reference: `data/ref/chrM.fa`. Input: `results/{sample}.bam`.
   - Write uncompressed VCF to a temporary path (e.g. `results/{sample}.vcf`); lofreq emits plain VCF.

7. **VCF compression and indexing**
   - Compress with `bgzip` (not `bcftools view -O z`) producing `results/{sample}.vcf.gz`.
   - Index with `tabix -p vcf results/{sample}.vcf.gz` → `results/{sample}.vcf.gz.tbi`.
   - Remove the intermediate uncompressed `.vcf`.

8. **Collapse step → `results/collapsed.tsv`**
   - For each sample, run `bcftools query -f '{sample}\t%CHROM\t%POS\t%REF\t%ALT\t%INFO/AF\n' results/{
   - Concatenate all four samples' output.
   - Prepend a single header line `sample\tchrom\tpos\tref\talt\taf` (tab-separated).
   - Output is tab-separated, one variant per line, header on, written to `results/collapsed.tsv`. Rebu

9. **Idempotency check**
   - Final pass: re-running the script on a fully populated `results/` exits 0, performs no work, and l

Plan v1.25 — plan/PLAN_v1p25.md

Per-sample mtDNA amplicon variant-calling plan

1. **Set globals and prepare results directory**
   - Define `THREADS=4` and the sample list: `M117-bl M117-ch M117C1-bl M117C1-ch`.
   - Create `results/` if missing. Use `set -euo pipefail`.
   - Treat every output step as idempotent: guard each artifact with an existence check (e.g. skip if `r

2. **Reference indexing (once, in `data/ref/`)**
   - `samtools faidx data/ref/chrM.fa` → produces `chrM.fa.fai`.
   - `bwa index data/ref/chrM.fa` → produces the `.amb .ann .bwt .pac .sa` set.
   - Skip both if the index files already exist.

3. **Per-sample alignment with `bwa mem`**
   - Use `bwa mem -t 4` with the paired FASTQs `data/raw/{sample}_1.fq.gz` and `data/raw/{sample}_2.fq.
   - Pass the read group via `-R` as a single double-quoted argument containing literal backslash-t bet
     - exact form: `-R "@RG\tID:{sample}\tSM:{sample}\tLB:{sample}\tPL:ILLUMINA"`
     - The `\t` must remain the two characters backslash and `t` - bwa parses them itself. Do NOT use `
     - Separators between key and value are colons `:`, not `=`.

4. **SAM → sorted BAM**
   - Pipe `bwa mem` stdout into `samtools sort -@ 4 -o results/{sample}.bam`.
   - Do NOT run `markdup` or `rmdup`: this is amplicon data where PCR duplicates are expected and biolo

5. **BAM indexing**
   - `samtools index -@ 4 results/{sample}.bam` → `results/{sample}.bam.bai`.

6. **Variant calling with `lofreq call-parallel`**
   - Exact command:
     ```
     ```

```
    lofreq call-parallel --pp-threads 4 -f data/ref/chrM.fa -o results/{sample}.vcf results/{sample}.b
    ```
```
   - Output: `results/{sample}.vcf` (uncompressed). lofreq emits plain VCF.

7. **VCF compression and indexing**
   - Compress with `bgzip` (not `bcftools view -O z`) producing `results/{sample}.vcf.gz`.
   - Index with `tabix -p vcf results/{sample}.vcf.gz` → `results/{sample}.vcf.gz.tbi`.
   - Remove the intermediate uncompressed `.vcf`.

8. **Collapse step → `results/collapsed.tsv`**
   - For each sample, run `bcftools query -f '{sample}\t%CHROM\t%POS\t%REF\t%ALT\t%INFO/AF\n' results/{
   - Concatenate all four samples' output.
   - Prepend a single header line `sample\tchrom\tpos\tref\talt\taf` (tab-separated).
   - Output is tab-separated, one variant per line, header on, written to `results/collapsed.tsv`. Rebu

9. **Idempotency check**
   - Final pass: re-running the script on a fully populated `results/` exits 0, performs no work, and l

**Plan v1.5 — plan/PLAN_v1p5.md**

```
Implementation Plan: Per-sample mtDNA Variant Calling

Boilerplate (top of `run.sh`)

```
set -euo pipefail
THREADS=4
SAMPLES=("M117-bl" "M117-ch" "M117C1-bl" "M117C1-ch")
mkdir -p results
```

All per-sample steps run in `for sample in "${SAMPLES[@]}"; do ... done`.

---

1. Reference indexing - BWA

```
bwa index data/ref/chrM.fa
```

2. Reference indexing - samtools faidx

```
samtools faidx data/ref/chrM.fa
```

3. Per-sample alignment + sort (one pipeline)

```
bwa mem -t 4 -R "@RG\tID:{sample}\tSM:{sample}\tLB:{sample}\tPL:ILLUMINA" data/ref/chrM.fa data/raw/{sa
```

4. BAM index
```
```

````
```
samtools index -@ 4 results/{sample}.bam
```

## 5. Variant calling - LoFreq

```
lofreq call-parallel --pp-threads 4 -f data/ref/chrM.fa -o results/{sample}.vcf results/{sample}.bam
```

## 6. VCF compression + tabix index

```
bgzip -f results/{sample}.vcf
```
```
tabix -p vcf results/{sample}.vcf.gz
```

## 7. Collapsed TSV

```
printf 'sample\tchrom\tpos\tref\talt\taf\n' > results/collapsed.tsv
```
```
bcftools query -f '%CHROM\t%POS\t%REF\t%ALT\t%INFO/AF\n' results/{sample}.vcf.gz | awk -v s={sample} 'B
```
````

Plan v1g — plan/PLAN_v1g.md

````
# Per-sample mtDNA amplicon variant-calling plan
<!--
v1g = v1 with Galaxy-IUC-derived CLI snippets injected per step where IUC has
clean coverage. Extracted mechanically by scripts/galaxy_to_snippet.py from
tools-iuc commit 39e745658a6ff7f013788871916574117a0f47f1 (2026-04-27).

IUC coverage map:
  bwa, bwa-mem        : extraction yields mostly noise (heavy macro use) - fallback to v1 prose
  samtools_faidx       : all-conditional command block - fallback to v1 prose
  samtools_sort        : partial extraction with placeholders - fallback to v1 prose
  samtools_index       : not in IUC - fallback to v1 prose
  lofreq_call_parallel : clean extraction - INJECTED (step 6)
  bgzip / tabix        : not in IUC - fallback to v1 prose
  bcftools_query       : format string in stripped Cheetah var - fallback to v1 prose
-->

1. **Set globals and prepare results directory**
   - Define `THREADS=4` and the sample list: `M117-bl M117-ch M117C1-bl M117C1-ch`.
   - Create `results/` if missing. Use `set -euo pipefail`.
   - Treat every output step as idempotent: guard each artifact with an existence check (e.g. skip if `

2. **Reference indexing (once, in `data/ref/`)**
   - `samtools faidx data/ref/chrM.fa` → produces `chrM.fa.fai`.
````

- `bwa index data/ref/chrM.fa` → produces the `.amb .ann .bwt .pac .sa` set.
- Skip both if the index files already exist.

3. **Per-sample alignment with `bwa mem`**
   - Use `bwa mem -t 4` with the paired FASTQs `data/raw/{sample}_1.fq.gz` and `data/raw/{sample}_2.fq.
   - Pass the read group via `-R` as a single double-quoted argument containing literal backslash-t bet
     - exact form: `-R "@RG\tID:{sample}\tSM:{sample}\tLB:{sample}\tPL:ILLUMINA"`
     - The `\t` must remain the two characters backslash and `t` - bwa parses them itself. Do NOT use `
     - Separators between key and value are colons `:`, not `=`.

4. **SAM → sorted BAM**
   - Pipe `bwa mem` stdout into `samtools sort -@ 4 -o results/{sample}.bam`.
   - Do NOT run `markdup` or `rmdup`: this is amplicon data where PCR duplicates are expected and biolo

5. **BAM indexing**
   - `samtools index -@ 4 results/{sample}.bam` → `results/{sample}.bam.bai`.

6. **Variant calling with `lofreq call-parallel`**
   - Galaxy IUC canonical invocation (extracted from `tools/lofreq/lofreq_call.xml` @ tools-iuc 39e7456
     ```
     lofreq call-parallel --pp-threads 4 --verbose
     --ref data/ref/chrM.fa --out results/{sample}.vcf
     --sig
     --bonf
     results/{sample}.bam
     ```
     (The bare `--sig` and `--bonf` lines come from Galaxy-runtime-supplied values; you can omit them a

7. **VCF compression and indexing**
   - Compress with `bgzip` (not `bcftools view -O z`) producing `results/{sample}.vcf.gz`.
   - Index with `tabix -p vcf results/{sample}.vcf.gz` → `results/{sample}.vcf.gz.tbi`.
   - Remove the intermediate uncompressed `.vcf`.

8. **Collapse step → `results/collapsed.tsv`**
   - For each sample, run `bcftools query -f '{sample}\t%CHROM\t%POS\t%REF\t%ALT\t%INFO/AF\n' results/{
   - Concatenate all four samples' output.
   - Prepend a single header line `sample\tchrom\tpos\tref\talt\taf` (tab-separated).
   - Output is tab-separated, one variant per line, header on, written to `results/collapsed.tsv`. Rebu

9. **Idempotency check**
   - Final pass: re-running the script on a fully populated `results/` exits 0, performs no work, and l

Plan v2 (detailed) — plan/PLAN.md

```
# Implementation Plan: Per-sample mtDNA Variant Calling

## Boilerplate (top of `run.sh`)
- First line after shebang: `set -euo pipefail`.
- Constants: `THREADS=4` and `SAMPLES=("M117-bl" "M117-ch" "M117C1-bl" "M117C1-ch")`.
- Create output dir: `mkdir -p results`.
- All per-sample steps must be wrapped in `for sample in "${SAMPLES[@]}"; do ... done`.

---
```

1. Reference indexing - BWA

```
bwa index data/ref/chrM.fa
```

- Outputs (5 sibling files): `data/ref/chrM.fa.amb`, `.ann`, `.bwt`, `.pac`, `.sa`.
- Idempotency guard: `[[ -f data/ref/chrM.fa.bwt ]] || bwa index data/ref/chrM.fa`
- Gotcha: `bwa index` writes outputs next to the input; the dir must be writable. No flags needed for a

2. Reference indexing - samtools faidx

```
samtools faidx data/ref/chrM.fa
```

- Output: `data/ref/chrM.fa.fai`.
- Guard: `[[ -f data/ref/chrM.fa.fai ]] || samtools faidx data/ref/chrM.fa`

3. Per-sample alignment + sort (one pipeline)

```
bwa mem -t 4 -R "@RG\tID:{sample}\tSM:{sample}\tLB:{sample}\tPL:ILLUMINA" data/ref/chrM.fa data/raw/{sa
```

- Output: `results/{sample}.bam`.
- Guard: `[[ -f results/{sample}.bam ]] || { bwa mem ... | samtools sort ... ; }` - wrap the whole pipe
- RG string gotchas (CRITICAL):
  - Use colons (`ID:`, `SM:`, `LB:`, `PL:`) - never `=`.
  - Use the **literal two characters** `\t` (backslash + t) inside the double-quoted string. Do NOT use
  - The whole `-R` value must be a single double-quoted argument.
- `samtools sort` trailing `-` reads from stdin.

4. BAM index

```
samtools index -@ 4 results/{sample}.bam
```

- Output: `results/{sample}.bam.bai`.
- Guard: `[[ -f results/{sample}.bam.bai ]] || samtools index -@ 4 results/{sample}.bam`
- Do NOT run `markdup` - this is amplicon data; PCR duplicates are expected and informative.

5. Variant calling - LoFreq

```
lofreq call-parallel --pp-threads 4 -f data/ref/chrM.fa -o results/{sample}.vcf results/{sample}.bam
```

- Output: `results/{sample}.vcf` (uncompressed).
- Guard: `[[-f results/{sample}.vcf || -f results/{sample}.vcf.gz]] || lofreq call-parallel --pp-thre
  (Check both because step 6 will delete the `.vcf` and leave `.vcf.gz`.)
- Gotchas: BAM is positional, NOT behind `-b`/`-i`. The flag is `--pp-threads`, not `-t` or `--threads

6. VCF compression + tabix index

```
bgzip -f results/{sample}.vcf
```
```
tabix -p vcf results/{sample}.vcf.gz
```

- Outputs: `results/{sample}.vcf.gz` and `results/{sample}.vcf.gz.tbi`.
- Combined guard: `[[-f results/{sample}.vcf.gz.tbi]] || { bgzip -f results/{sample}.vcf && tabix -p
- Gotchas: `bgzip` operates **in place** - it deletes `results/{sample}.vcf` after writing `.vcf.gz`.

7. Collapsed TSV (rebuild every run)

Do NOT guard this step - always overwrite, since per-sample VCFs may have changed.

Header (overwrite):

```
printf 'sample\tchrom\tpos\tref\talt\taf\n' > results/collapsed.tsv
```

Per sample, append:

```
bcftools query -f '%CHROM\t%POS\t%REF\t%ALT\t%INFO/AF\n' results/{sample}.vcf.gz | awk -v s={sample} 'B
```

- Gotchas:
  - The format string uses `%INFO/AF`, not `%AF` - bcftools requires the `INFO/` prefix for INFO fields
  - The `\t` and `\n` inside `-f '...'` are **bcftools format codes**, parsed by bcftools itself; keep
  - awk's `OFS="\t"` is required so `print s,$0` joins with a tab (`$0` already contains the tabbed bcf
  - Use `>` for the header line, `>>` for every per-sample append.

Idempotency summary
- Steps 1-6 each have a `[[ -f <sentinel> ]] ||` guard on their final output. A second invocation on a
- Step 7 is intentionally rebuilt from scratch on every run (header `>`, then append per sample). This

Plan v2_defensive — plan/PLAN_v2_defensive.md

`run.sh` Implementation Plan - chrM amplicon variant calling

0. Script preamble (top of file)

1. First line after shebang: `set -euo pipefail`.
2. Constants: `THREADS=4` and `SAMPLES=("M117-bl" "M117-ch" "M117C1-bl" "M117C1-ch")`.
3. Paths: `REF=data/ref/chrM.fa`, `OUT=results`.
4. `mkdir -p "$OUT"`.
5. Initialize the failure log every run (truncate, no header - pure TSV body): `: > "$OUT/failures.log"
6. Track survivors with an array: `SURVIVORS=()` and a counter `OK=0`.

0a. Defensive helper `try`

Define this function exactly:

```
try() {  # try <sample> <step_label> <validation_cmd_string> -- <cmd...>
  local sample="$1" step="$2" validate="$3"; shift 3
  [[ "$1" == "--" ]] && shift
  if "$@" && eval "$validate"; then return 0; fi
  "$@" && eval "$validate" && return 0
  printf '%s\t%s\t%s\n' "$sample" "$step" "command_or_validation_failed" >> "$OUT/failures.log"
  return 1
}
```

Behavior: runs `cmd`, then evaluates the validation string; on any failure retries the **same cmd + sam

For reference-prep steps (no sample), use sample label `__ref__` and `exit 1` instead of `continue` on

---

1. Reference preparation (once, before the sample loop)

1a. `bwa index`

```
bwa index data/ref/chrM.fa
```

- Outputs: `data/ref/chrM.fa.{amb,ann,bwt,pac,sa}`.
- Idempotency guard: `[[-f data/ref/chrM.fa.bwt]] || try __ref__ bwa_index '[[-s data/ref/chrM.fa.bw
- On failure after retry: `echo "[run.sh] reference index failed" >&2; exit 1`.

1b. `samtools faidx`

```
samtools faidx data/ref/chrM.fa
```

- Output: `data/ref/chrM.fa.fai`.
- Guard: `[[-f data/ref/chrM.fa.fai]] || try __ref__ faidx '[[-s data/ref/chrM.fa.fai]]' -- samtools
- On failure: `exit 1`.

---

2. Per-sample loop

`for s in "${SAMPLES[@]}"; do ... done`. Inside the loop, every `try` failure must `continue` to the ne

Step 2a - Align + sort → `results/{s}.bam`

Exact pipeline (the RG string must be a literal double-quoted string containing the four characters `\`

```
```

```
bwa mem -t 4 -R "@RG\tID:${s}\tSM:${s}\tLB:${s}\tPL:ILLUMINA" \
  data/ref/chrM.fa data/raw/${s}_1.fq.gz data/raw/${s}_2.fq.gz \
  | samtools sort -@ 4 -o results/${s}.bam -
```

- Output: `results/${s}.bam`.
- Guard: `[[ -f results/${s}.bam ]] && samtools quickcheck results/${s}.bam` → skip; else run via `try
- Validation string passed to `try`: `'samtools quickcheck results/'"$s"'.bam'`.
- Wrap the whole pipeline in a tiny inline shell function or `bash -c` because `try` takes argv, not a
- Gotchas: literal `\t` only; PL is `ILLUMINA` (uppercase); `samtools sort`'s trailing `-` reads stdin

Step 2b - BAM index → `results/{s}.bam.bai`

```
samtools index -@ 4 results/${s}.bam
```

- Output: `results/${s}.bam.bai`.
- Guard: `[[ -s results/${s}.bam.bai ]]` → skip.
- Validation: `'[[ -s results/'"$s"'.bam.bai ]]'`.
- No duplicate marking (amplicon data).

Step 2c - Variant calling → `results/{s}.vcf`

Exact command (no substitutions, no extra flags):

```
lofreq call-parallel --pp-threads 4 -f data/ref/chrM.fa -o results/${s}.vcf results/${s}.bam
```

- Output: `results/${s}.vcf` (uncompressed; lofreq writes plain VCF here).
- Guard: skip if `results/${s}.vcf.gz` already exists and tabix index validates (step 2d covers it); ot
- Validation: `'[[ -s results/'"$s"'.vcf ]] && bcftools view -h results/'"$s"'.vcf > /dev/null'`.
- Gotcha: `--pp-threads` is mandatory for `call-parallel`; `-t` is wrong here.

Step 2d - Compress + tabix → `results/{s}.vcf.gz` + `.tbi`

Two separate invocations, each wrapped in its own `try`:

```
bgzip -f results/${s}.vcf
tabix -p vcf results/${s}.vcf.gz
```

- Outputs: `results/${s}.vcf.gz`, `results/${s}.vcf.gz.tbi`.
- Guard for the pair: `[[-s results/${s}.vcf.gz && -s results/${s}.vcf.gz.tbi]] && bcftools view -h r
- Validation after `bgzip`: `'[[-s results/'"$s"'.vcf.gz]] && bcftools view -h results/'"$s"'.vcf.gz
- Validation after `tabix`: `'[[ -s results/'"$s"'.vcf.gz.tbi ]]'`.
- Gotcha: `bgzip -f` **deletes** `results/${s}.vcf` on success - that is expected; do not look for it a

Step 2e - Mark survivor

After all four steps succeed for sample `s`: `SURVIVORS+=("$s"); OK=$((OK+1))`.

---

3. Collapsed TSV (after the loop, only over `SURVIVORS`)

1. Write header (always, overwrite):

```
printf 'sample\tchrom\tpos\tref\talt\taf\n' > results/collapsed.tsv
```

2. For each `s` in `"${SURVIVORS[@]}"`:

```
bcftools query -f '%CHROM\t%POS\t%REF\t%ALT\t%INFO/AF\n' results/${s}.vcf.gz \
  | awk -v s=${s} 'BEGIN{OFS="\t"}{print s,$0}' >> results/collapsed.tsv
```

Wrap in `try "$s" collapse '[[ -s results/collapsed.tsv ]]' -- ...` so a single broken VCF only loses t

---

4. Final summary + exit code

1. Build a comma-separated list of failed samples by `cut -f1 results/failures.log | sort -u | grep -v
2. Identify the first failing step per failed sample (e.g. `awk -F'\t' '!seen[$1]++{print $1" failed at
3. Emit on **stderr**, as the very last line:

```
[run.sh] <OK>/<TOTAL> samples completed; <sample> failed at step <label> - see results/failures.log
```

If `OK == ${#SAMPLES[@]}`: `[run.sh] 4/4 samples completed; no failures`.

4. Exit policy: `if (( OK >= 1 )); then exit 0; else exit 1; fi`.

---

5. Idempotency note

Every step is guarded by an `[[ -f ... ]] && validation` check before invoking `try`. A second run on a